Beamer$^+$

Interactive Presentations

A Guided Tour

press → to begin

what's inside

Beamer$^+$ turns a static PDF into a live, interactive lecture

every tile below has a demo later in this deck, so press → to explore


video & audio


3D models

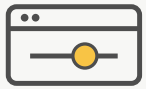

split view

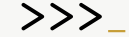

reveal steps


ink & shapes



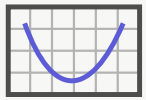
magnifier


live widgets



Python


live polls


scratch paper

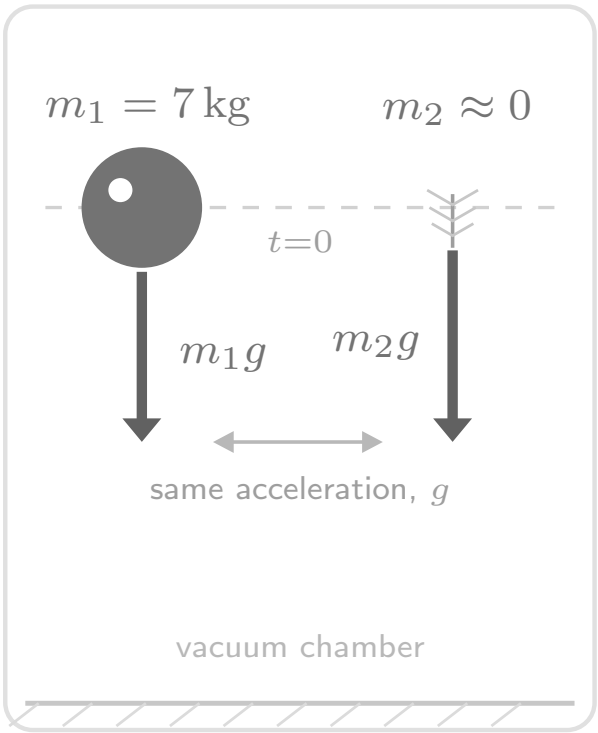

screen recording


notes PDF

videos/audio

supported formats include `.mp4`, `.mov`, `.mpeg`, `.mp3`, `.wav`

example usage: demonstrating Newton's 2nd law in a Physics course



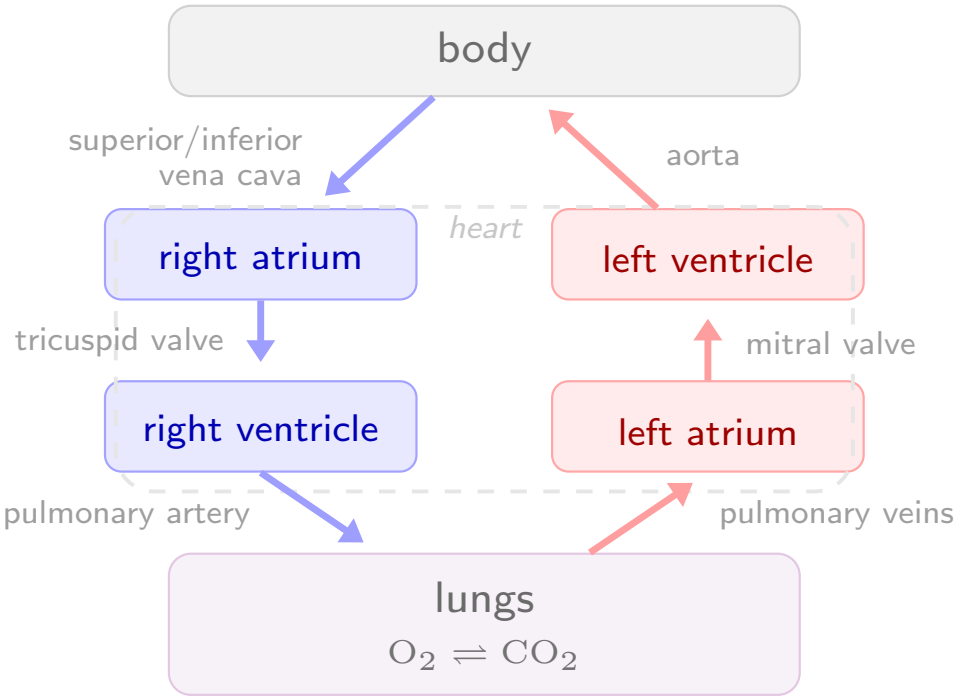
gravitational acceleration is independent of mass

try clicking!

models

supported formats include `.glpb`

example usage: explaining blood flow through the heart in an Anatomy course



blood flow through the heart
(deoxygenated blood / oxygenated blood)

try rotating!

reveal steps

uncover a slide one piece at a time, with no \pause or recompiling

example usage: walking through a proof in a Discrete Math class

Claim. $\sqrt{2}$ is irrational.

1. Suppose not: $\sqrt{2} = \frac{p}{q}$ with p, q integers in *lowest terms*.
2. Squaring, $p^2 = 2q^2$, so p^2 is even $\Rightarrow$ p is even: $p = 2k$.
3. Then $4k^2 = 2q^2$, so $q^2 = 2k^2$ $\Rightarrow$ q is even too.
4. So p and q are both even, and share a factor of 2.

Contradiction! Lowest terms can't both be even. $\square$

press → to reveal!

ink & annotations

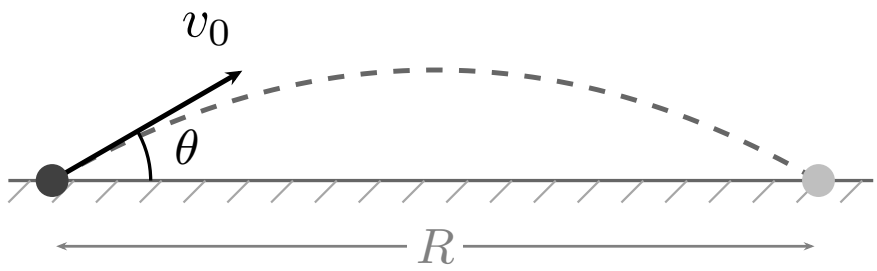
pens, highlighters, shapes, LaTeX text boxes, a spotlight and a magnifier

example usage: working a projectile problem live in a Physics class

A ball is kicked at $v_0 = 20\,\mathrm{m/s}$, at $\theta = 30^\circ$ above level ground.
How far away does it land?

The range is

$$R = \frac{v_0^2 \sin 2\theta}{g} = \frac{(20)^2 \sin 60^\circ}{9.81}$$



Psst, you found the fine print! Scroll while the magnifier is on to zoom anywhere from 1.5x to 5x. It works on ink, videos and widgets too.

try the magnifier on the fine print!

split view

allows two slides to be shown simultaneously

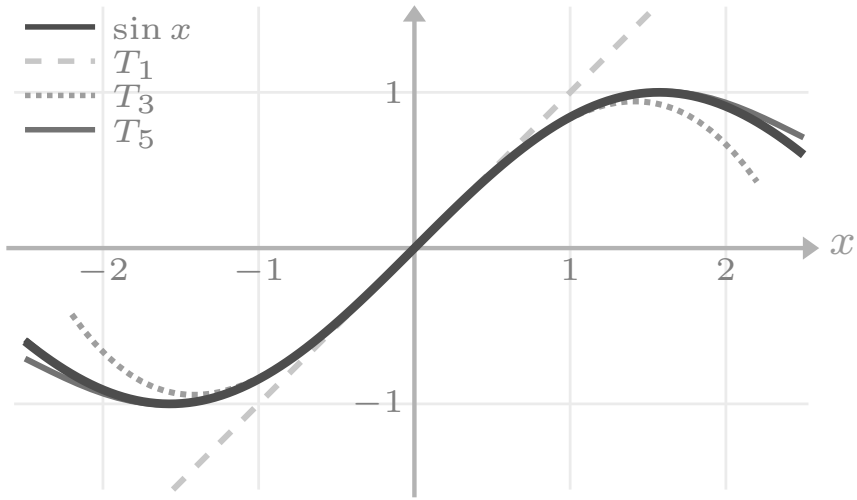
example usage: explaining Taylor series in a Calculus class

Any smooth function f can be approximated around $x_0 = 0$ as:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

y





expansion of $f(x) = \sin(x)$ around $x_0 = 0$

function plotter

2D curves and 3D surfaces you can rotate, zoom and edit live

example usage: classifying critical points in a Multivariable Calculus class

At a critical point, $\nabla f = \mathbf{0}$. The *second-derivative test* looks at

$$D = f_{xx} f_{yy} - f_{xy}^2.$$

For $f(x, y) = a\,x^2 - y^2$ at the origin:

$D = (2a)(-2) - 0^2 = -4a$

$a > 0$: a **saddle point**

$a < 0$: a local maximum

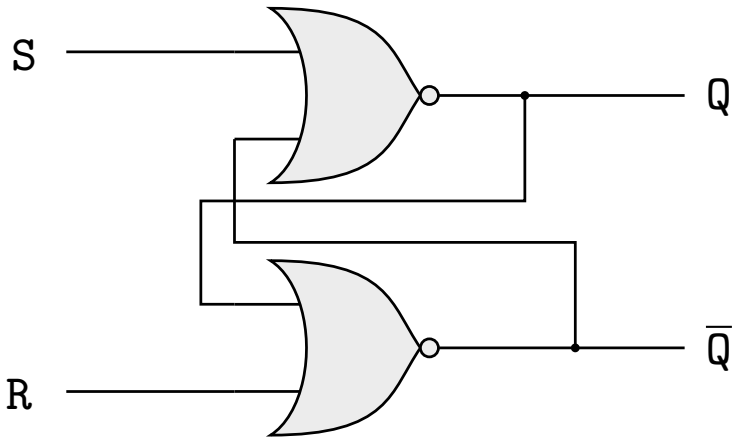
try dragging the slider for a!

widgets

supported formats include `.html`

example usage: explaining latches in a Computer Organization class

A **latch** stores one bit of state using cross-coupled gate feedback.



cross-coupled NOR gates
inputs are *active-high*

S	R	Q	$\overline{Q}$	state
0	0	Q	$\overline{Q}$	hold
1	0	1	0	set
0	1	0	1	reset
1	1	0	0	forbidden

truth table for an SR latch
implemented with NOR gates

record & share

the lecture doesn't have to stay in the room

example usage: posting the recording and annotated notes right after class



record the tab

slides, ink, widgets and sound, saved every second so a crash loses nothing



notes PDF

one handout with your ink, every widget as it stands, and the poll results



one ZIP file

slides, media, widgets, ink and poll setup together, ready to reopen next term



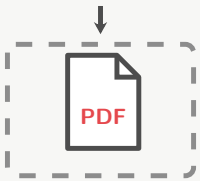
try the record button!

make your own

start from the PDF you already have

everything you've seen in this deck takes three steps

1



load your slides

drop a PDF onto the window, or a Beamer$^+$ ZIP you saved earlier

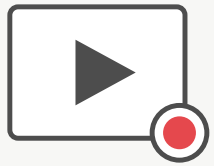
2



add the extras

open edit mode to place videos, 3D models, widgets and polls

3



present

draw, poll and record live, then download the ZIP and your notes